

Julien Soulé

Doctor in Computer Science
AI & Multi-Agent Systems

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Professional Profile

Doctor in Computer Science specializing in Multi-Agent Systems, Multi-Agent Reinforcement Learning (MARL), and neuro-symbolic AI for cyberdefense and operational decision support. I bridge research-grade AI methods with deployable architectures through R&D experience with **Thales Land & Air Systems**, **Thales Cyber Solutions**, the **University of Luxembourg**, and public-sector partners. My work emphasizes robust autonomous systems, human-in-the-loop supervision, explainability, simulation, and operational constraints in safety-critical environments.

Technical and R&D Interests

MARL; Multi-Agent System Organizations; Neuro-Symbolic AI; Cyberdefense & Intelligent Autonomous Agents; Simulation, World Models & Digital Twins; Safety-critical and Adversarial Environments; Software Architecture for MAS; Explainability and Human-in-the-loop AI.

Appointments

- 2026–present **Postdoctoral Researcher**, *University of Luxembourg*, Luxembourg
Research on neuro-symbolic approaches for multi-agent systems, MARL, world models, and applied AI for industrial and public-sector projects, including collaborations with **Thales Cyber Solutions** and the Luxembourgish police.
- 2022–2025 **Doctoral Researcher**, *Thales Land & Air Systems + Université Grenoble Alpes*, France
Design of multi-agent architectures for cyber-resilient and safety-critical defense systems (**Autonomous Intelligent Cyberdefense Agents – AICA**). Application of MARL under organizational constraint. Development of simulation platforms for adversarial cyber scenarios.
- 2021–2022 **Research Engineer**, *Thales Land & Air Systems*, Rennes, France
Multi-agent modeling of cyber environments and autonomous anomaly detection in Thales LAS context.
- 2020–2021 **Software & Research Engineer**, *Atos*, Toulouse, France
Software engineering for satellite command and control systems for **Centre National d'Etudes Spatiales** and the **European Space Agency**.
- 2020 **Software Engineering Internship**, *Atos*, Toulouse, France
Development of the **ISIS (CNES Initiative for Space Innovative Standards)** system for satellite command & control (Python, Bash, KVM, Grafana, Django).
- 2019 **Software Engineering Internship**, *SQLI*, Toulouse, France
Cybersecurity, maintenance, and software engineering tasks for **Airbus Helicopters** projects.

Education

- 2022–2025 **PhD in Computer Science**, *Université Grenoble Alpes (UGA)*, France
Dissertation defended on **17 November 2025**. Title: “*On the Organization of a Cyberdefense Multi-Agent System*”. Status: **Degree conferred**. Supervisors: Jean-Paul Jamont, Michel Occello, Louis-Marie Traonouez, Paul Théron.

- 2019 **Exchange Semester in Computer Engineering**, *École de Technologie Supérieure (ETS)*, Montréal, Canada
- 2015–2021 **Integrated Bachelor's and Master's Degree**, *INSA Rennes*, France
Specialization in Computer Science and Cybersecurity.

Research & Industrial Projects

ALIAS Applied R&D project within **Thales Cyber Solutions** for External Attack Surface Management (EASM). The platform combines reinforcement learning, MARL, world models, large language models, and neuro-symbolic AI in a human-in-the-loop framework for reliable, explainable, and adaptive cyber defense.

AssistDiFo Applied R&D project with the Luxembourg Police to assist investigators in analyzing large volumes of heterogeneous digital evidence. The platform uses named entity recognition, retrieval-augmented generation (RAG), and large language models while keeping investigators in control of the workflow.

CybMASDE A research platform combining MARL, organizational modeling, and cyberdefense simulation to support the design and analysis of intelligent multi-agent architectures (with application for **Thales**-related projects).

MOISE+MARL A proof-of-concept framework integrating organizational roles, missions, and goals into MARL environments to guide learning and improve agent coordination.

Selected Publications

Journal Articles

Full publication list available on request or via DBLP.

- **(JAAMAS 2026)** – *Assisting Multi-Agent System Design with MOISE+ and MARL: The MAMAD Method* (Soulé et al.).
- **ROIA (accepted, 2025)** – *MOISE+MARL: un cadre organisationnel pour l'explicabilité et le contrôle en apprentissage par renforcement multi-agent* (Soulé et al.).

International Conferences and Workshops

- **MASSpace / AAMAS Workshops 2026** – *Toward Automated Operational Assist in Satellite Fleet Management via Organizational MARL* (Soulé).
- **AAMAS 2025** – *An Organizationally-Oriented Approach to Enhancing Explainability and Control in Multi-Agent Reinforcement Learning* (Soulé et al.).
- **IEEE CLOUD 2025** – *Streamlining Resilient Kubernetes Autoscaling with Multi-Agent Systems via an Automated Online Design Framework* (Soulé et al.). DOI: 10.1109/CLOUD67622.2025.00015.
- **AIAI 2024** – *A MARL-Based Approach for Easing MAS Organization Engineering* (Soulé et al.).

Invited Talks

- **Mar. 2023** — *Invited talk for the NATO **Cyb'Air** chair. Thales, Dassault, École de l'air et de l'espace.*

Awards & Distinctions

- 2025 Best Paper Award – JFSMA 2025, for work on explainability and control in multi-agent reinforcement learning.

Additional Activities

- 2023–present **Treasurer** of the **AICA International Work Group**.
- 2024–present Conference reviewing in particular for **AAMAS**, **ICAART**, **IEEE SMC**, **ICDL**, and **CNSM**.

Skills

AI & Learning	RL/MARL, Neuro-Symbolic AI, LLMs and RAG, Learning under Constraints, Behavior Analysis and Interpretability
Multi-Agent Systems	Agent Architectures, Coordination and Cooperation, Organizational Models, Distributed Decision-Making, Simulation, World Models and Digital Twins
Systems & Engineering	Python, Gym/PettingZoo, JAX / JaxMARL, PyTorch, Optuna, RLlib, MARLlib, JS/HTML/CSS, Flask, Docker, Linux, Kubernetes, HPC Platforms
Applied Domains	Embedded Autonomous Systems, Drone Swarms (IoT/IoBT), Cyberdefense, Adversarial and Multi-Agent Environments
Languages	French (native), English (professional), Spanish (intermediate), Catalan (intermediate), Japanese (basic)